

YT-RAGBot

Semantic Video Intelligence

Project Architecture & Tech Stack

Large Language Model:	Azure OpenAI (GPT-4o)
RAG Framework:	LangChain (LCEL Orchestration)
Search Engine:	FAISS (Semantic Similarity Search)
Backend Services:	FastAPI (High-Performance API)
Data Extraction:	yt-dlp, youtube-transcript-api

What It Does

YT-RAGBot is an advanced AI system designed to bridge the gap between passive video consumption and active knowledge acquisition. By transforming YouTube videos into interactive, high-fidelity knowledge bases, it enables users to engage with content through natural language queries.

Key Features:

- Triple-Redundancy Ingestion: Ensures near-100% transcript fetch success rate.
- Temporal Citations: Responses include [MM:SS] timestamps for factual verification.
- Intelligent Chunking: Preserves context across 1000-character semantic segments.
- Isolated Indexing: Vector stores are partitioned by video ID for sub-second retrieval.

Explore the Source Code on GitHub

<https://github.com/SwayamAg/YT-RAG-Bot-Semantic-Video-Intelligence>